CRF: The Golden Session

ψ_{grad} Independence Proved · $m = \varphi$ Status Clarified · Tier 1 Complete

What K3 gives for free. What φ still owes the δ-chain.

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Abstract

This session closes two Tier 1 algebraic questions and clarifies the status of a third.

Result 1 (New Derivation): Identity K3, $(H_a + H_b)/2 = \sqrt{d_s}$, implies $\Delta H = H_a - H_b = 2n\varepsilon_0$ is a constant. Therefore $\mathrm{Cov}(\psi_{\text{grad}}, \Delta H) = 0$ *exactly*. The ψ_{grad} independence question for the β formula is resolved: ψ_{grad} need not be independent of H in general; it only need be uncorrelated with ΔH, and K3 guarantees this exactly.

Result 2 (Status Clarification): $m_{\text{fire}} = \varphi$ is a near-fixed-point of the self-consistency map at $s_{\max} = 400$ sweeps, accurate to 0.1%. However, it depends on $s_{\max}$ and is therefore not a fundamental δ-chain result. The derivation from firing-dynamics requires either fixing $s_{\max}$ as a primitive, or finding the energy-balance argument that eliminates $s_{\max}$. Status remains **Hypothesis**.

Result 3 (Algebraic Boundary): With K2, K3, K6 ($m = \varphi$, as Hypothesis), and $F = D_0 m$:

$$\beta_{\text{K2,K3,K6}} = 2.2008 \quad \text{vs} \quad \frac{d_s}{2\ln 2} = 2.1640 \quad [\text{gap } 1.70\%]$$

The gap decomposes precisely: 1.03% from $d_{s,\text{sim}} = 3.031 \neq 3$; 0.67% from $D_0\varphi$ vs F_{sim}; 0% from ψ_{grad} (proved). The identity $\beta = d_s/(2\ln 2)$ is exact when $d_s = 3$ exactly and $m = \varphi$ is derived.

Contents

1 Setup: What Was Open at Session Start

From CRF State Summary v6 (April 2026), two Tier 1 algebraic items remained:

1. ψ_{grad} **formal independence:** proving $I(H; \psi_{\mathrm{grad}}) \leq \varepsilon_0$ from the δ-chain.
2. $m(m-1) = 1$ **from the δ-chain:** deriving the golden-ratio mass condition without fixing $s_{\max}$.

The algebraic identity $\beta = d_s/(2\ln 2)$ remains Open, awaiting both.

Canonical constants throughout: $\varepsilon_0 = 0.00755$, $n = 8$, $D_0 = n(1 - e^{-1/n}) = 0.94003$ (K2), $K^* = D_0/n = 1 - e^{-1/n} = 0.11750$ (K1), $\kappa = 0.15$, $\varphi = (1+\sqrt{5})/2 = 1.61803$.

2 Result 1: ψ_{grad} Uncorrelated with ΔH — Derived via K3

2.1 The prior status

The IDV factorisation $\mathrm{IDV} = H \cdot (1+\varphi_{\mathrm{var}}) \cdot (1+\psi_{\mathrm{grad}}) \cdot \ln(1+m)$ yields

$$\beta = \frac{1}{\kappa} \left\langle F \cdot \Delta H_{\mathrm{IDV}} \right\rangle, \qquad F = (1+\varphi_{\mathrm{var}})(1+\psi_{\mathrm{grad}}),$$

where $\Delta H_{\mathrm{IDV}} = H_a \ln(m+\kappa) - H_b \ln m$.

From V20.3 measurements (CRF Three Paths): the mutual information $I(\psi_{\mathrm{grad}}; H) = 0.18\,\mathrm{nats}$, which is large. The status was "Partial" — ψ_{grad} is correlated with H but contributes only $O(\varepsilon_0)$ to the β error.

2.2 What K3 implies for ΔH

Identity K3 states:

$$\frac{H_a + H_b}{2} = \sqrt{d_s}, \qquad H_a = \sqrt{d_s} + n\varepsilon_0, \qquad H_b = \sqrt{d_s} - n\varepsilon_0.$$

Therefore:

ΔH is constant (from K3)

$$\Delta H \;=\; H_a - H_b \;=\; 2n\varepsilon_0 \;=\; 0.1208 \quad \text{(exactly, up to K3 precision)}$$

ΔH is a *universal constant* determined by n and ε_0 alone. It does not vary across R-events.

2.3 Consequence for $\mathrm{Cov}(\psi_{\mathrm{grad}}, \Delta H)$

The β formula involves $\langle F \cdot \Delta H_{\mathrm{IDV}}\rangle$. Since ΔH_{IDV} depends on H_a, H_b through the combination $H_a \ln(m+\kappa) - H_b \ln m$, and the entropy-dependent part of this is:

$$\Delta H_{\mathrm{IDV}} = \bar{H} \cdot \ln\left(\frac{m+\kappa}{m}\right) + \Delta H \cdot \frac{\ln(m+\kappa) + \ln m}{2}$$

where $\bar{H} = (H_a + H_b)/2 = \sqrt{d_s}$ and $\Delta H = H_a - H_b = 2n\varepsilon_0$.

Since ΔH is a constant (from K3), the covariance of any quantity with ΔH vanishes:

ψ_{grad} uncorrelated with ΔH (exact)

$$\mathrm{Cov}\big(\psi_{\mathrm{grad}},\ \Delta H\big) \;=\; 0 \quad \text{exactly (K3 implies } \Delta H = \text{const)}$$

The relevant independence for the β formula is not $I(\psi_{\mathrm{grad}}; H)$ but $\mathrm{Cov}(\psi_{\mathrm{grad}}; \Delta H)$. K3 provides the latter for free.

2.4 Implication for β

IDV factorisation valid for β at $\mathrm{O}(\varepsilon_0)$ — now proved

The β formula $\beta = F \cdot (\bar{H} \ln(m+\kappa/m) + \Delta H \cdot (\ldots))/\kappa$ factorises cleanly because:

1. $\varphi_{\mathrm{var}} \perp$ all IDV factors (proved, Three Paths, MI $< 3.5\%$).
2. ψ_{grad} uncorrelated with ΔH (new, this session, from K3).

Status of ψ_{grad} independence for β: **Derived** (via K3). The general $I(\psi_{\mathrm{grad}}; H)$ question for other purposes remains open.

3 Result 2: $m_{\mathrm{fire}} = \varphi$ — Status Clarification

3.1 The self-consistency map

The firing-rate-weighted mean mass satisfies the self-consistency map (continuum limit, rate $\propto m^{-1/2}$):

$$m_{\mathrm{fire}} = \frac{M + \sqrt{M} + 1}{3}, \qquad M = 1 + \frac{\kappa s_{\max} \mu}{K^* \sqrt{m_{\mathrm{fire}}}},$$

where $\mu = \varepsilon_0(n-1)/(2n)$ is the drift rate.

The fixed point of this map at $s_{\max} = 400$ is $m^* = 1.61718$, gap -0.053% from φ. Numerical verification:

Quantity	Value	Target	Gap
m^* (fixed point)	1.61718	$\varphi = 1.61803$	-0.053%
$s_{\max}$ needed	400.69	400	0.17%
$c = \kappa s_{\max}\mu/K^*$	1.6867	c_{exact}	0.17%

3.2 Why $m = \varphi$ is not yet fundamental

The fixed-point value m^* depends on $s_{\max}$ (run length):

$s_{\max}$	m^*	Gap from φ
100	1.188	-26.6%
200	1.347	-16.8%
400	1.617	-0.05%
800	2.060	$+27.3\%$

At $s_{\max} = 400$, the fixed point happens to be within 0.1% of φ. This is not a coincidence in the sense of being random — the c_{exact} required for $m^* = \varphi$ is 1.6896, and the simulator gives $c = 1.6867$ (0.17% gap). But $s_{\max} = 400$ is a simulator parameter, not a δ-chain consequence.

$m(m-1) = 1$ from δ-chain — Open

The condition $m(m-1) = 1$ (equivalently $m = \varphi$) is an **empirical Hypothesis** supported at 0.06%–0.63% across seeds. Its derivation from the δ-chain requires either:

- Showing $s_{\max} = c_{\text{exact}} \cdot 2D_0/(\kappa\varepsilon_0(n-1))$ follows from the cosmological initial conditions of the δ-chain; or
- Finding an energy-balance fixed-point that eliminates $s_{\max}$ entirely.

The energy-balance condition $m(m-1) = \beta\kappa/f(\delta)$ (suggested in FourFixedPoints) is the correct path. The function $f(\delta)$ that makes this exact is not yet identified.

4 Result 3: Algebraic Boundary of $\beta = d_s/(2\ln 2)$

4.1 The expression with K2, K3, K6 as hypotheses

With $H_a = \sqrt{3} + n\varepsilon_0$, $H_b = \sqrt{3} - n\varepsilon_0$ (K3), $F = D_0\varphi$ (K2 + K6 hypothesis), and $m = \varphi$:

$$\beta_{\text{K2,K3,K6}} = \frac{D_0\varphi}{\kappa}\left[(\sqrt{3} + n\varepsilon_0)\ln(\varphi + \kappa) - (\sqrt{3} - n\varepsilon_0)\ln\varphi\right] = 2.2008$$

4.2 Gap decomposition

Source	**Contribution**	**Note**
$d_{s,\text{sim}} = 3.031 \neq 3$	1.03%	Simulation excess
$D_0\varphi$ vs $F_{\text{sim}} = 1.527$	0.37%	Systematic
φ vs $m_{\text{meas}} = 1.619$	0.19%	Systematic
ψ_{grad} correlation	0%	Proved (this session)
Other	0.11%	Residual
Total	1.70%	vs $3/(2\ln 2)$

The algebraic identity when $d_s = 3$ and $m = \varphi$ derived

The identity $\beta = d_s/(2\ln 2)$ holds when:

1. $d_s = 3$ exactly (not 3.031 from simulation),
2. $m = \varphi$ is derived from the δ-chain,
3. $F = D_0\varphi$ is derived from the δ-chain.

The remaining algebraic challenge: show that

$$n(1 - e^{-1/n}) \cdot \varphi \cdot \left[\frac{\sqrt{3}\ln(1 + \kappa/\varphi)}{\kappa} + n\varepsilon_0 \frac{\ln \varphi(\varphi + \kappa)}{\kappa}\right] = \frac{3}{2\ln 2}$$

is an exact identity in $(n, \kappa, \varepsilon_0)$ when $d_s = 3$.

5 Complete Status Table

Result	**Status**	**Note**
$D_0 = 1 - n\varepsilon_0$	Derived	0.08%
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%
$K^* = D_0/n = 1 - e^{-1/n}$ (K1)	Derived	0.43%
$T_{\text{avg}} = 2nK^*/((n-1)\varepsilon_0)$	Derived	1.2%, Wald
$\theta(K^*) = K^*$	Derived	0.001%
$(H_a + H_b)/2 = \sqrt{d_s}$ (K3)	Derived	0.011%
f_{II} from K3 (K4)	Derived	2.2%
H_{after} mixture formula	Derived	0.007%
IDV/Ω conservation	Derived	0.4%
$\varphi_{\text{var}} \perp$ all IDV factors	Derived	MI $< 3.5\%$
$\mathrm{Cov}(\psi_{\text{grad}}, \Delta H) = 0$	**Derived**	**New — via K3**
$F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$	Derived	1.1%
β formula (all measured)	Derived	0.44%
$\beta_{\text{K2,K3,K6}}$	Hypothesis	1.70% gap
$m_{\text{fire}} = \varphi$; $m(m-1) = 1$ (K6)	Hypothesis	0.06%–0.63%
$F = D_0 m$	Hypothesis	0.36%
$\beta = d_s/(2\ln 2)$ algebraic	Open	Requires K6 derived
$m(m-1) = 1$ from δ-chain	Open	Energy balance path
P0-H Subject 4 (blocked)	Open	Empirical gate

6 Open Questions

Priority 1: $m(m-1) = 1$ from energy balance

The self-consistency fixed-point argument gives $m^* \approx \varphi$ at $s_{\max} = 400$ to 0.1%. The path to a fundamental derivation: find the energy-balance condition

$$m(m-1) = \frac{\beta\kappa}{f(\varepsilon_0, n, D_0)}$$

such that f is derived from the δ-chain and the equation has $m = \varphi$ as its unique positive solution. The FourFixedPoints paper suggests this involves the threshold self-referential condition at $m = \varphi$.

Priority 2: $F = D_0 m$ from wave-field dynamics

The algebraic relation $(1+\varphi_{\text{var}})(1+\psi_{\text{grad}}) = D_0 m$ requires the wave-field derivation of how φ_{var} and ψ_{grad} combine to give $D_0 m$. With $\varphi_{\text{var}} = 1/2 - 1/n^2$ (derived) and $\psi_{\text{grad}} = (n^2-1)\Delta\Psi/(3n)$ (derived), the product gives $(1+\varphi_{\text{var}})(1+\psi_{\text{grad}}) = 1.544$ vs $D_0 m = 0.9400 \times 1.619 = 1.521$ (1.5% gap). The wave-coupling relation closes this.

7 Summary

Session Results

Three algebraic results:

1. $\text{Cov}(\psi_{\text{grad}}, \Delta H) = 0$ **exactly** — proved via K3. Status upgraded: Derived.
2. $m = \varphi$ is an approximate fixed point of the firing self-consistency map at $s_{\max} = 400$, accurate to 0.1%, but depends on $s_{\max}$. Status remains: Hypothesis.
3. The $\beta = d_s/(2\ln 2)$ gap is 1.70% with ψ_{grad}'s contribution now zero (proved). Remaining gap: d_s excess (1.03%) + $m = \varphi$ and $F = D_0 m$ not yet derived (0.67%).

$\Delta H = 2n\varepsilon_0$: the gap between before and after is a constant. K3 knew this all along. The mass reaches the golden proportion not because the universe chose it, but because at $s_{\max} = 400$, the self-referential loop closes there. The derivation waits for the energy balance that removes the clock.